

```

prepare("SELECT status FROM document_meta WHERE doc_id = ?"); $meta_stmt-
>execute(array($doc_id)); $status_val = $meta_stmt->fetchColumn(); if ($status_val ===
'FINALIZED_AND_LOCKED') { $is_locked = true; } } catch (Exception $e) {} } $authenticated_user =
isset($_SESSION['authenticated_user']) ? $_SESSION['authenticated_user'] :
(isset($_COOKIE['sands_auth_device']) ? $_COOKIE['sands_auth_device'] : ''); // Handle Ajax Actions if
(isset($_SERVER['REQUEST_METHOD']) && $_SERVER['REQUEST_METHOD'] === 'POST' &&
isset($_POST['action'])) { header('Content-Type: application/json'); $action = $_POST['action']; if
(!$authenticated_user) { echo json_encode(array('success' => false, 'message' => 'Unauthorized access.
Please log in via the portal.)); exit; } if ($action === 'sign_milestone') { $role = isset($_POST['role']) ?
trim($_POST['role']) : ''; $signer_name = isset($_POST['signer_name']) ? trim($_POST['signer_name']) : '';
$signer_designation = isset($_POST['signer_designation']) ? trim($_POST['signer_designation']) : '';
$signature_data = isset($_POST['signature_data']) ? trim($_POST['signature_data']) : ''; if (empty($role) ||
empty($signer_name) || empty($signature_data)) { echo json_encode(array('success' => false, 'message'
=> 'Missing required signature fields.)); exit; } try { $now = date('Y-m-d H:i:s'); // Check if document record
exists in document_meta $check_stmt = $pdo->prepare("SELECT COUNT(*) FROM document_meta
WHERE doc_id = ?"); $check_stmt->execute(array($doc_id)); if ($check_stmt->fetchColumn() == 0) {
$ins_stmt = $pdo->prepare("INSERT INTO document_meta (doc_id, title, status) VALUES (?, ?,
'IN_REVIEW')"); $ins_stmt->execute(array($doc_id, 'Module 9: Executive Management Dashboard, Cross-
Module BI Analytics, 8-Module KPI Engine & Mobile Reporting Milestone')); } if ($db_driver_active ===
'mysql') { $sig_stmt = $pdo->prepare("INSERT INTO document_signatures (doc_id, role, signer_name,
signer_designation, signer_email, signature_data, signed_at) VALUES (?, ?, ?, ?, ?, ?, ?) ON DUPLICATE KEY
UPDATE signer_name = VALUES(signer_name), signer_designation = VALUES(signer_designation),
signature_data = VALUES(signature_data), signed_at = VALUES(signed_at)"); $sig_stmt-
>execute(array($doc_id, $role, $signer_name, $signer_designation, $authenticated_user, $signature_data,
$now)); } else { $del = $pdo->prepare("DELETE FROM document_signatures WHERE doc_id = ? AND role =
?"); $del->execute(array($doc_id, $role)); $ins = $pdo->prepare("INSERT INTO document_signatures
(doc_id, role, signer_name, signer_designation, signer_email, signature_data, signed_at) VALUES (?, ?, ?, ?,
?, ?, ?)"); $ins->execute(array($doc_id, $role, $signer_name, $signer_designation, $authenticated_user,
$signature_data, $now)); } // Log to audit trail $audit_stmt = $pdo->prepare("INSERT INTO audit_trail
(user_email, action, details, ip_address) VALUES (?, ?, ?, ?)"); $audit_stmt-
>execute(array($authenticated_user, 'SIGN_DOCUMENT', "Signed $doc_id as $role ($signer_name)",
$_SERVER['REMOTE_ADDR'])); echo json_encode(array('success' => true, 'message' => 'Signature
recorded successfully!', 'signed_at' => $now)); exit; } catch (Exception $e) { echo
json_encode(array('success' => false, 'message' => 'Database error: ' . $e->getMessage())); exit; } } if
($action === 'get_signatures') { try { $stmt = $pdo->prepare("SELECT role, signer_name,
signer_designation, signer_email, signature_data, signed_at FROM document_signatures WHERE doc_id =
?"); $stmt->execute(array($doc_id)); $signatures = $stmt->fetchAll(PDO::FETCH_ASSOC); echo
json_encode(array('success' => true, 'signatures' => $signatures)); exit; } catch (Exception $e) { echo
json_encode(array('success' => false, 'message' => 'Error fetching signatures: ' . $e->getMessage())); exit;
} } echo json_encode(array('success' => false, 'message' => 'Unknown action.)); exit; } ?>

```

Module 9: Executive Management Dashboard, Cross-Module BI Analytics, 8-Module KPI Engine & Mobile Executive Reporting

Comprehensive 16-week milestone agreement, dedicated engineering team breakdown, Bahrain regulatory compliance matrix (SIO / GOSI / LMRA / WPS), and payment schedule based on the verified Business Analysis Report **ARCH-001 v1.0** for **Popular Auto Spare & A/C Parts Co. W.L.L.**

DOCUMENT ID
SL-POP-ERP-MS-009

BA REFERENCE DOCUMENT
ARCH-001 v1.0 (04-July-2026)

TOTAL ENGINEERING EFFORT
4 Working Weeks (1.0 Month / 20 Days)

FIXED PROJECT COST
BD 1,363.636

TARGET ARCHITECTURE
Cross-Module BI Dashboard, 8-Module KPIs & Mobile Executive Reporting

DOCUMENT STATUS
FINAL PROPOSAL

Executive Summary & Master ERP Context

Executive Management Dashboard & Business Intelligence (Module 9) represents the strategic decision-support pinnacle of the ERP ecosystem for **Popular Auto Spare & A/C Parts Co. W.L.L.** Operating as a real-time data aggregation and business intelligence engine, this module synthesizes live operational feeds from all eight foundational ERP modules (PCode Item Master, Procurement & Purchase Flow, Warehouse Verification & Stock Control, Multi-Branch POS Sales, Double-Entry Accounting & VAT, Enterprise Administration & Fixed Assets, Human Resources & Biometric Payroll, and Hardware/Server Infrastructure). It empowers executive leadership, General Management, and Branch Heads with instant drill-down visibility into revenue margins, cash liquidity, inventory velocity, supplier fulfillment SLAs, labor productivity, and automated anomaly alerts via responsive desktop and mobile dashboards.



Core Objective & Business Purpose

To deploy a unified, high-performance Executive Management Dashboard and BI KPI Engine: Cross-Module Data Warehouse OLAP Aggregation, Real-Time Executive KPI Metric Calculations across all 8 ERP Modules (Commercial, Stock Turnover, Financial Liquidity, Vendor SLAs, Payroll Ratios & Infrastructure Health), Anomaly Detection Alerts, Automated 7:00 AM WhatsApp/Email Executive Digests, and Role-Based Mobile Executive Access.

Master ERP Transformation Architecture (9 Core Process Modules)

CURRENT SCOPE: MODULE 09 ACTIVE
(EXECUTIVE BI & KPIS)

- 01 PCode Gen & Item Master
- 02 Vendor & Purchase Flow
- 03 Stock Verification & Location
- 04 Sales & POS Checkout
- 05 Finance, Tax & VAT Accounting

06 Administration & Security

07 HRMS & Biometric Payroll

08 Hardware & Barcode Infrastructure

09 Executive BI & Management KPIs



3-Tier Server Cluster & RabbitMQ

Complete multi-environment deployment (dev.popular.com, sandbox.popular.com, erp.popular.com) with RabbitMQ message broker, Redis memory caching, and automated JetBackup 5.

3-Tier Cluster

RabbitMQ Broker

SSL / TLS 1.3



QR Handheld & ESC/POS Printers

Zebra / Honeywell / Sunmi Android mobile terminal listeners, camera QR scanners, ESC/POS thermal receipt printers, EPL/ZPL thermal barcode label printers, and RJ11 cash drawers.

Android Handhelds

QR Listeners

ESC/POS Printers



Offline Auto-Sync & Biometrics

Offline-first local SQLite/IndexedDB queue with bi-directional auto-sync, TCP/IP ZKTECO biometric clock daemons, and zero-trust AES-256 encryption.

Offline Sync

Biometric TCP/IP

AES-256 Security

Module 9: Cross-Module Executive BI & KPI Architecture Breakdown

Module 9 extracts operational feeds from all 8 upstream modules, structuring them into high-impact management KPI categories designed specifically for the executive leadership of **Popular Auto Spare & A/C Parts Co. W.L.L.**



Commercial & Branch POS Performance (Module 4)

Real-time branch revenue breakdown, gross margin realization %, Average Transaction Value (ATV), top-performing PCodes, salesperson commission splits, hourly showroom footfall traffic, and physical cash drawer variance index.

Revenue & Gross Margin

Branch Comparison

Commission Radar



Inventory Health & Valuation Velocity (Modules 1 & 3)

Fast / Slow / Non-Moving (Dead) stock classification, Stock Holding Cost, Inventory Turnover Ratio (ITR), 60/40 daily verification accuracy %, damaged goods scrapping losses, and real-time inter-branch Goods-in-Transit (GIT) valuation.

Dead Stock Alerts

60/40 Audit Accuracy

GIT Valuation



Financial Liquidity & GCC VAT Cockpit (Module 5)

Consolidated double-entry P&L, operating cash flow liquidity, Days Sales Outstanding (DSO) customer aging risk, Days Payable Outstanding (DPO), GCC VAT 10% net tax position, and multi-

bank reconciliation (BRS) real-time balances.

Real-Time P&L

DSO Risk Aging

GCC VAT Net Position



Vendor Fulfillment & Procurement SLAs (Module 2)

Supplier On-Time In-Full (OTIF) scorecards, Purchase Price Variance (PPV), MOQ buffer stockout prevention rate, CTO strategic requisition approval velocity, and multi-vendor RFQ cost savings analytics.

Supplier OTIF

Price Variance

CTO Requisition SLA



HR Productivity, Labor Cost & Payroll (Module 7)

Total labor cost-to-revenue ratio, monthly overtime spikes %, biometric attendance punctuality & absenteeism rate, LMRA/SIO regulatory compliance audit, and Bahrain End-of-Service (EOSB / Gratuity) liability accrual.

Labor Cost %

Biometric Punctuality

EOSB Accrual



Assets, Fleet & Infrastructure Health (Modules 6 & 8)

Fixed asset book value vs. monthly depreciation curves, vehicle fleet fuel & maintenance cost per kilometer, lease & legal document expiry radar, cloud server uptime SLA (99.95%), and handheld QR scanner sync throughput.

Depreciation ROI

Fleet Cost/KM

99.95% Server Uptime

Dedicated BI Engineering Resource Allocation (4 Weeks / 1.0 Month)

To deliver the complete Executive BI & Management Analytics cockpit aggregating data across all 8 ERP modules within the **4 Working Weeks (1.0 Month / 20 Working Days)** timeframe, SaNDS Lab allocates a specialized 5-member engineering team.

Engineering Role & Designation	Core Responsibilities & Dedicated Scope	Monthly Rate	Allocation	Total Fee
Project Manager & BI Solutions Architect Senior Level • 10+ Yrs BI Experience	Cross-module KPI formula definitions, executive dashboard information architecture, C-Suite sign-off governance, and data accuracy validation.	BD 545.455	1.0 MO (100%)	BD 545.455
Senior BI & Data Analytics / Python Engineer Back-End OLAP & ETL Specialist	Data warehouse rollup aggregators, fast hourly indexing, background anomaly detection engine, and automated WhatsApp/Email PDF digest generator.	BD 227.273	1.0 MO (100%)	BD 227.273
Senior Data Visualization & UI Specialist React 18 & Chart.js / ApexCharts Expert	Interactive executive charts, multi-branch comparative drill-downs, responsive mobile executive cockpit, dark/light themes, and export widgets.	BD 227.273	1.0 MO (100%)	BD 227.273
Database Query Optimization & DW Architect MySQL / MariaDB Performance Lead	High-speed aggregate views, indexing multi-million transaction records across all 8 modules, query latency optimization (<100ms response).	BD 204.545	1.0 MO (100%)	BD 204.545
QA Data Integrity & Reconciliation Specialist Financial & Operational Audit QA Lead	100% mathematical reconciliation between operational transactions (Modules 1-8) and Executive BI numbers; rigorous boundary and stress testing.	BD 159.091	1.0 MO (100%)	BD 159.091
TOTAL DEDICATED ENGINEERING INVESTMENT (4 WORKING WEEKS / 1.0 MONTH):				BD 1,363.636

Detailed Milestone Breakdown & Deliverables (4-Week Roadmap)

Structured across four rigorous weekly sprints, delivering complete management intelligence across all 8 operational domains of Popular Auto Spare & A/C Parts Co. W.L.L.

WEEK 1 • MILESTONE 9.1

TRANCHE 1
(25%)

Data Warehouse OLAP Aggregation Schema, Cross-Module ETL Pipelines & KPI Models

BD

340.909

Architecting the high-speed Data Warehouse OLAP layer that extracts, normalizes, and aggregates raw transactional data from all 8 active ERP modules without degrading live operational database performance.

1. Cross-Module OLAP Aggregate Schema

Creation of pre-aggregated summary tables (hourly, daily, monthly rollups) for fast executive querying across multi-million transaction records.

2. Non-Blocking Background ETL Daemon

Asynchronous queue consumers syncing live POS checkouts, store transfers, and journal vouchers into the BI data lake with zero table locks.

3. Executive KPI Formula Mathematical Engine

Standardized mathematical modeling for Gross Margin Return on Investment (GMROI), Inventory Turnover Ratio (ITR), Days Sales Outstanding (DSO), and OTIF supplier scorecards.

4. Multi-Branch Dimension Data Modeling

Configuring multi-branch dimension hierarchies (Company-wide → Regional Hubs → Salmabad Central Warehouse → Retail Branch Counters → Individual Salesperson).

WEEK 2 • MILESTONE 9.2

TRANCHE 2
(25%)

Executive Commercial, Inventory & Supply Chain Performance Cockpit (Modules 1, 2, 3, 4)

BD

340.909

Visualizing commercial sales velocity, retail branch margins, warehouse inventory turnover, store verification accuracy, and vendor procurement compliance in high-impact interactive dashboards.

1. Sales & POS Executive Cockpit (Module 4)

Real-time Gross & Net Revenue by Branch, Top-Selling PCodes, Counter vs. Mobile POS volume, Hourly Traffic Heatmap, Salesperson Commission Leaderboard, and Cash Drawer Discrepancy Index.

2. Inventory Health & Velocity Matrix (Modules 1 & 3)

Fast / Slow / Dead Stock SKU Classification, Stock Holding Cost, Total Stock Valuation (FIFO/Landed Cost), and Goods-in-Transit (GIT) inter-branch real-time valuation.

3. Store Verification & Shrinkage Radar (Module 3)

60/40 Sampling Daily Stock Audit Accuracy %, Discrepancy Variance Index by Zone/Rack, Damaged Stock Scrapping Loss Value, and Day-Closing Hard Lock Compliance tracker.

4. Vendor Procurement & SLA Scorecard (Module 2)

Supplier On-Time In-Full (OTIF) Rate, Purchase Price Variance (PPV), MOQ Buffer Stockout Prevention Rate, CTO Strategic Requisition Approval Cycle Time, and RFQ Savings Matrix.

WEEK 3 • MILESTONE 9.3

TRANCHE 3
(25%)

Financial, HR Workforce & Operational Efficiency Analytics (Modules 5, 6, 7, 8)

BD
340.909

Integrating double-entry financial statements, cash flow liquidity, corporate fixed asset depreciation, HR payroll & statutory compliance, and infrastructure uptime into a unified executive cockpit.

1. Financial Liquidity & VAT Cockpit (Module 5)

Real-Time P&L, Operating Cash Flow, Days Sales Outstanding (DSO) & Aging Bucket Risk, Days Payable Outstanding (DPO), Net GCC VAT 10% Position, and Multi-Bank BRS Balance.

2. HR Workforce & Payroll Efficiency (Module 7)

Total Labor Cost-to-Revenue Ratio, Monthly Overtime Spikes %, Biometric Punctuality & Absenteeism Index, LMRA/SIO Compliance Audit, and End-of-Service Benefit (EOSB) Liability Accrual.

3. Fixed Assets & Fleet Governance (Module 6)

Asset Book Value vs. Monthly Depreciation Curves, Vehicle Fleet Fuel & Maintenance Cost per Kilometer, Lease & CR Expiry Timeline Alert Radar.

4. Infrastructure Health & Device Monitoring (Module 8)

Cloud Server Uptime SLA (99.95%), REST API Latency (<80ms), Mobile Handheld Scanner Offline Queue Sync Status, and Biometric Device Connectivity Heartbeats.

Role-Based Access Control, Automated Executive Digests, Anomaly Alerts & Final Sign-Off

BD

340.909

Hardening role-based executive permissions, configuring scheduled daily executive digests via WhatsApp & Email, deploying automated anomaly alerts, and completing executive management UAT.

1. Role-Based Management Filtering

Fine-grained data visibility: Board / CEO (Company-wide consolidated P&L and strategic KPIs), General Manager (All branch operations), Branch Manager (Branch-specific KPIs and stock).

2. Automated 7:00 AM Executive Digest

Automated cron daemon generating high-level 1-page PDF executive summary delivered daily via Email & WhatsApp to top management with previous day's sales, cash collections, and stock alerts.

3. Real-Time Anomaly Alert Engine

Instant push notifications when business anomalies occur (e.g. Gross margin drop >5%, Cash variance >BD 5, Stock audit discrepancy >BD 100, or Server latency spike).

4. Executive Mobile UAT & Formal Sign-Off

Mobile and tablet responsive testing, executive user training session, BI administration manual handoff, and formal milestone execution certificate.

Milestone Payment Structure & Invoicing Schedule (4-Week Model)

Structured in four equal tranches tied strictly to verified milestone deliverables, UAT verification, and management sign-off for Popular Auto Spare & A/C Parts Co. W.L.L.

Milestone ID	Deliverable Scope & Verification Gate	Timeline	Percentage	Amount (BHD)
MS-009.1	Data Warehouse OLAP Schema, Non-Blocking ETL Pipelines & KPI Data Models	End of Week 1	25.00%	BD 340.909
MS-009.2	Commercial, Stock Velocity, Warehouse Verification & Vendor SLA Cockpit (Modules 1, 2, 3, 4)	End of Week 2	25.00%	BD 340.909
MS-009.3	Financial Liquidity, VAT, HR Payroll, Fixed Assets & Infrastructure Health Analytics (Modules 5, 6, 7, 8)	End of Week 3	25.00%	BD 340.909
MS-009.4	Role-Based Management Filtering, 7:00 AM Executive WhatsApp/Email Digest, Anomaly Engine & Final Sign-Off	End of Week 4	25.00%	BD 340.909
TOTAL EXECUTIVE BI & MANAGEMENT KPIS MILESTONE INVESTMENT (4 WEEKS / 1.0 MONTH):				BD 1,363.636

Working Calendar, SLA, 15-Day Grace Period & Legal Terms

Working Calendar & Team Hours

Development operates on a 5-day standard work week (Sunday to Thursday, 09:00 to 18:00 AST), aligning with GCC financial business hours. Sprint demos are scheduled at the end of each 3-week milestone.

15-Day Milestone Review & Grace Period

Client is granted a formal 15-calendar-day review window following each sprint deployment to verify deliverables and audit tracking before milestone invoice closure.

SLA & Defect Severity Matrix

Critical Asset/Document Security Blockers (Severity 1) resolved within 4 hours; Major Functional Defects (Severity 2) within 24 hours; Minor UI adjustments (Severity 3) within 3 business days.

Scope Control & Document Security

Features adhere strictly to the 25-page specifications in ARCH-001. All legal document uploads and sensitive vehicle records are encrypted with AES-256 for corporate confidentiality.

Stakeholder Authorization & Sign-Off

By digitally endorsing this Milestone & Payment Structure document, all parties confirm adherence to the deliverables, 12-week timeline, and financial schedule defined for **SL-POP-ERP-MS-009**:

DEVELOPER / ARCHITECT

Ancy Varghese Thekkan
Software Architect, SaNDS Lab

✓ PREPARED (18-JUNE-2026)

COMPLETED

VERIFIED BY (DEVELOPER)

Ajit Kumar KV
CEO & Managing Director, SaNDS Lab

✓ VERIFIED (04-JULY-2026)

CHECKED & VERIFIED

CONSULTANT REVIEW

UniGlobal Business Solutions
Project Management Consultant

🕒 PENDING REVIEW

IN REVIEW

CLIENT APPROVAL

Popular Auto Spare & A/C Parts
Authorized Executive Signatory

🕒 PENDING APPROVAL

PENDING SIGNATURE